Junwang Zhao

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Education

Peking University Sep, 2013 – Jul, 2016

Master of Computer Architecture

Nanjing University of Science and Technology Sep, 2009 – Jul, 2013

Bachelor of Computer Science

Career

Ant Group Platform Technology Group

Senior Database Kernel Developer Oct, 2024 – now

Transwarp Technology Infrastructure Department

Senior Database Kernel Developer Jul, 2023 – Sep, 2024

Alibaba Cloud Database OLAP Department

Senior Database Kernel Developer May, 2021 – June, 2023

Huawei IT Product Line

Senior developer Apr, 2018 – May, 2021

Projects

GraphUniverse Graph Database Oct, 2024 – now

Own optimizer and vector index kernel development for GraphUniverse, improving complex graph query performance and extensibility.

- Designed and implemented a graph query/execution optimization framework (MLIR-based) for critical complex query paths.
- Built a vector index abstraction layer to support unified integration and extension across multiple vector index libraries.
- Advanced Full Text Search support for graph data scenarios.
- Optimized multi-type node and variable-length path queries, improving executability and stability of complex graph statements.

Hippo Vector Database Jul, 2023 – Sep, 2024

Led Hippo vector database kernel development, focusing on index architecture abstraction, performance optimization, and engineering toolchain improvements.

- Abstracted vector index interfaces by modeling index data structures as a one-level LSM Tree, enabling extensibility and easier integration of diverse vector index libraries.
- Integrated DiskANN and optimized key paths, delivering a 30% overall performance improvement.
- Designed and implemented HippoBench with multiple open-source datasets, unifying tooling for POC tuning and performance benchmarking.

AnalyticDB PostgreSQL Iceberg Table May, 2021 – Jun, 2023

Led ADBPG integration with Apache Iceberg to connect data lake and data warehouse analytics workflows.

- Implemented iceberg-c SDK and iceberg-fdw.
- Enabled querying Iceberg tables on OSS to reduce data silos and connect lake/warehouse data.
- Drove architecture design and prototype validation for native-table-style Iceberg read/write support via table access method adaptation.
- Explored writing PostgreSQL logical replication changes into Iceberg tables, validating real-time sync and lakehouse ingestion paths.

FusionStorage OBS Index Layer Apr, 2018 – May, 2021

OBS Index Layer based on MongoDB, and utilizes the mongo-rocks storage engine to persist data to FusionStorage.

- Adapt MongoDB transport layer to use libxnet(a wrapper lib for ibverbs, RoCE, bypass kernel)
- Integrate the mongo-rocks storage engine with FusionStorage's PLOG interface.
- Hot spot analysis, improve write, query, and Compaction performance through reasonable batch processing, index optimization, and parameter tuning measures.

Open Source Contribution

- PostgreSQL (Significant Contributor)
  - Add fast path for foreign key constraint checks - Author - PG19
  - SQL Property Graph Queries (SQL/PGQ) - Main Reviewer - PG19
  - Introduce a SQL-callable function array_sort(anyarray) - Author - PG18
- Apache Iceberg C++ (Co-creator)
  - Apache Iceberg C++
- pgiceberg
  - Brings Apache Iceberg to Postgres and vice versa
- pgvectorBench
  - A lightweight, fast, flexible, and easy-to-use benchmarking tool dedicated to pgvector
- pg_cron
  - add possibility to schedule jobs on the last day of month
  - PRs contributed to pg_cron

Skills

- C/C++ > Rust > Python > Java
- Familiar with Epoll and Reactor Pattern
- Daily Used Tools: GDB/Perf/Valgrind/gperftools

Others

- Blog: https://zhjwpku.com
- Wechat Official Account: PostgreSQL Opensource Ecosystem
- English: CET-6 VET/VEWT
- Paper Notes(Chinese): paper notes
- GSoC: Google Summer of Code (X.Org, Ceph)